# Ensemble Methods

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# What are Ensemble methods?

- **Ensemble learning** is a machine learning paradigm where multiple models (often called **"weak learners"**) are trained to solve the same problem and combined to get better results.
- The main hypothesis is that when weak models are correctly combined, we can obtain more accurate and/or robust models.

# Ensemble methods?-Weak Learners

- In ensemble learning theory, we call **weak learners** (or base models) models that can be used as **building blocks** for designing more complex models by combining several of them.
- Most of the time, these basics models perform not so well by themselves either because they have **a high bias** (low degree of freedom models, for example) or because they have too much variance to be robust (high degree of freedom models, for example).
- Then, **the idea of ensemble methods** is **to try reducing bias and/or variance of such weak learners by combining several of them** together in order **to create a strong learner (or ensemble model)** that achieves better performances.

# Ensemble methods?-Weak Learners

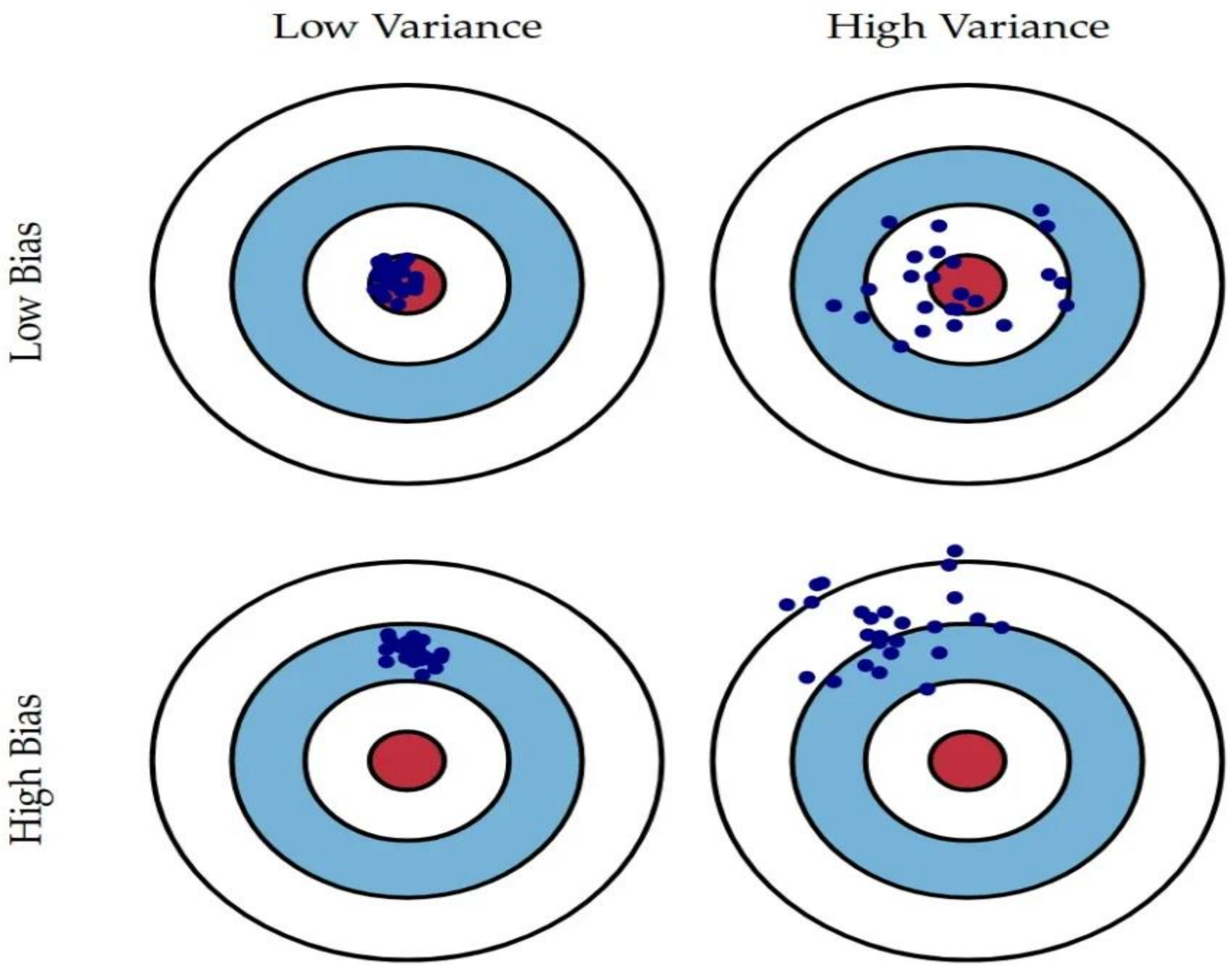



Fig. 1 Graphical illustration of bias and variance.

# Combine weak learners

- In order to set up an ensemble learning method, we first need to select our base models to be aggregated.
- Most of the time (including in the well-known bagging and boosting methods) **a single base learning algorithm** is used so that we **have homogeneous weak learners** that are trained in different ways.
- The ensemble model we obtain is then said to be **"homogeneous".**
- However, there also exist some methods that use **different type of base learning algorithms:** some heterogeneous weak learners are then combined into an **"heterogeneous ensembles model".**

# Combine weak learners

- **Three major kinds of meta-algorithms that are used for combining weak learners:**
  - **Bagging**, that often considers homogeneous weak learners, **learns them independently** from each other **in parallel** and combines them following some kind of deterministic averaging process
  - **Boosting**, that often considers homogeneous weak learners, **learns them sequentially in a very adaptative way** (**a base model depends on the previous ones**) and combines them following a deterministic strategy
  - **Stacking,** that often considers **heterogeneous weak learners**, **learns them in parallel** and combines them by training a meta-model to output a prediction based on the different weak models' predictions.

# Focus on bagging

- In **parallel methods** we fit the **different considered learners independently** from each others and, so, it is possible to train them concurrently.
- The most famous such approach is **“bagging”** (standing for **“bootstrap aggregating”**) that aims at producing an ensemble model that is more robust than the individual models composing it.

# Bootstrapping

❖ This statistical technique consists in generating samples of size B (called bootstrap samples) from an initial dataset of size N by randomly drawing with replacement B observations.

# Ensemble Learning

- Employ multiple learners and combine their predictions.
- Methods of combination:
  - Bagging, boosting, voting
  - Error-correcting output codes
  - Mixtures of experts
  - Stacked generalization
  - Cascading
  - …
- **Advantage:** improvement in predictive accuracy.
- **Disadvantage:** it is difficult to understand an ensemble of classifiers

# Why Do Ensembles Work?

**Dietterich(2002) showed that ensembles overcome three problems:**

- ***The Statistical Problem*** arises when the hypothesis space is too large for the amount of available data. Hence, there are many hypotheses with the same accuracy on the data and the learning algorithm chooses only one of them! There is a risk that the accuracy of the chosen hypothesis is low on unseen data!
- ***The Computational Problem*** arises when the learning algorithm cannot guarantee finding the best hypothesis.
- ***The Representational Problem*** arises when the hypothesis space does not contain any good approximation of the target class(es).

T.G. Dietterich, Ensemble Learning, 2002.

# Simple Ensemble Techniques

**There are Three Types of basic Techniques**

1. Max Voting
2. Averaging
3. Weighted Averaging

# Max Voting

The max voting method is generally used for classification problems.

In this technique, multiple models are used to make predictions for each data point. The predictions by each model are considered as a 'vote'.

The predictions which we get from the majority of the models are used as the final prediction.

For example, when you asked 5 of your colleagues to rate your movie (out of 5);

The result of max voting would be something like this:

**Example-**

| Colleague 1 | Colleague 2 | Colleague 3 | Colleague 4 | Colleague 5 |
|---|---|---|---|---|
| 5 | 4 | 5 | 4 | 4 |

**Final rating-4**

# Averaging

Similar to **the max voting technique**, multiple predictions are made for each data point in averaging.

In this method, we take an average of predictions from all the models and use it to make the final prediction.

**Averaging** can be used for making **predictions in regression problems** or while **calculating probabilities for classification problems**.

For example, in the below case, the averaging method would take the average of all the values.

i.e. $\mathbf{(5+4+5+4+4)/5 = 4.4}$

# Weighted Average

This is **an extension of the averaging method.**

**All models are assigned different weights** defining **the importance of each model for prediction.**

For instance, **if two of your colleagues are critics**, while others have no prior experience in this field, then **the answers by these two friends are given more importance as compared to the other people.**

The result is calculated as

[(**5*0.23**) + (**4*0.23**) + (5*0.18) + (4*0.18) + (4*0.18)] = **4.41.**

# Advanced Ensemble Techniques

**Advanced Ensemble Techniques are-**

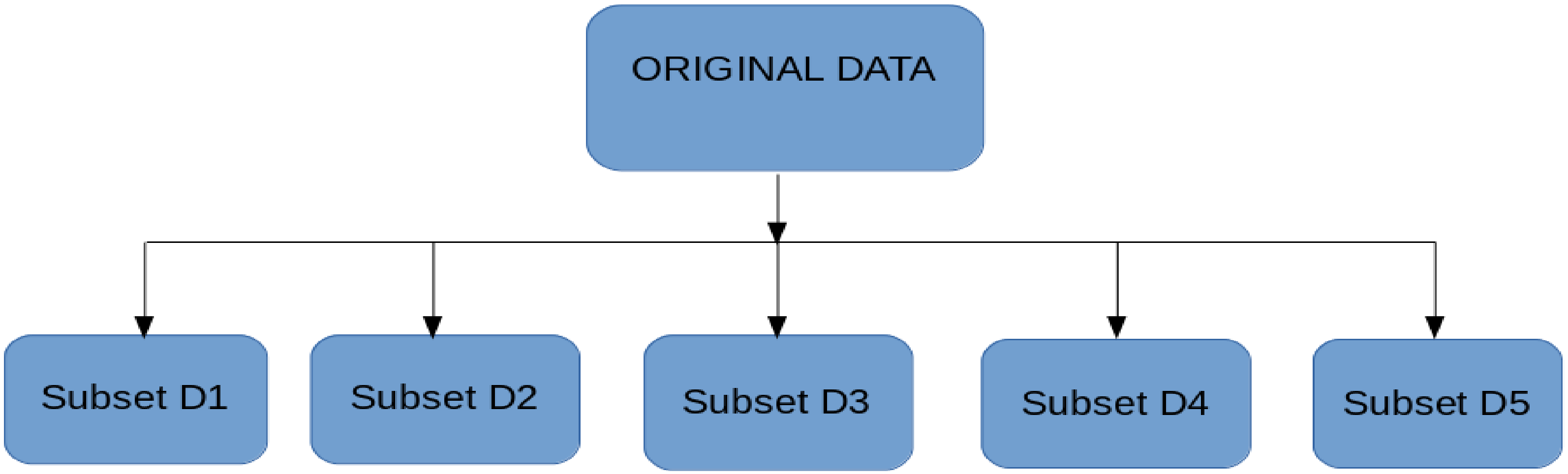
1. Bagging
2. Boosting
3. Stacking

# Bagging

The idea behind bagging is combining the results of multiple models (for instance, all decision trees) to get a generalized result.

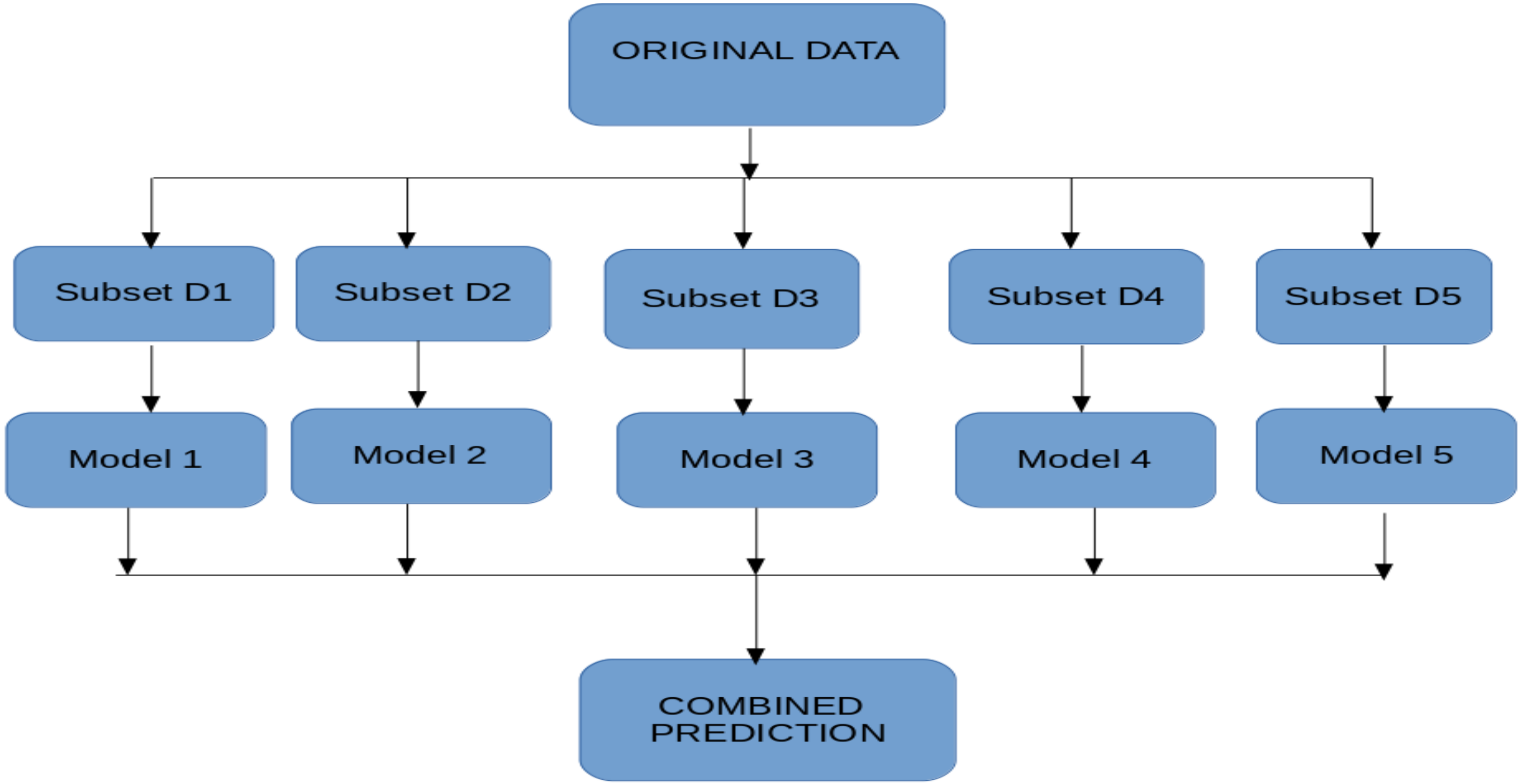
**Bootstrapping** is a sampling technique in which we create subsets of observations from the original dataset, with replacement. The size of the subsets is the same as the size of the original set.

**Bagging** (or Bootstrap Aggregating) technique uses these subsets (bags) to get a fair idea of the distribution (complete set). The size of subsets created for bagging may be less than the original set.



# Bagging

1. Multiple subsets are created from the original dataset, selecting observations with replacement.
2. A base model (weak model) is created on each of these subsets.
3. The models run in parallel and are independent of each other.
4. The final predictions are determined by combining the predictions from all the models



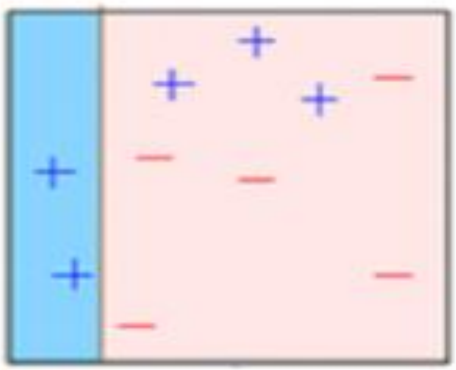
# Boosting

Before we go further, here’s another question for you: If a data point is incorrectly predicted by the first model, and then the next (probably all models), will combining the predictions provide better results? Such situations are taken care of by boosting.

**Boosting** is a sequential process, where each subsequent model attempts to correct the errors of the previous model.

The succeeding models are dependent on the previous model. Let’s understand the way boosting works in the below steps.

1. A subset is created from the original dataset.
2. Initially, all data points are given equal weights.
3. A base model is created on this subset.
4. This model is used to make predictions on the whole dataset



# Boosting

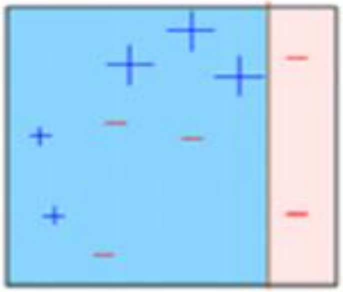
5.Errors are calculated using the actual values and predicted values.

6.The observations which are incorrectly predicted, are given higher weights.

(Here, the three misclassified blue-plus points will be given higher weights)

7. Another model is created and predictions are made on the dataset.

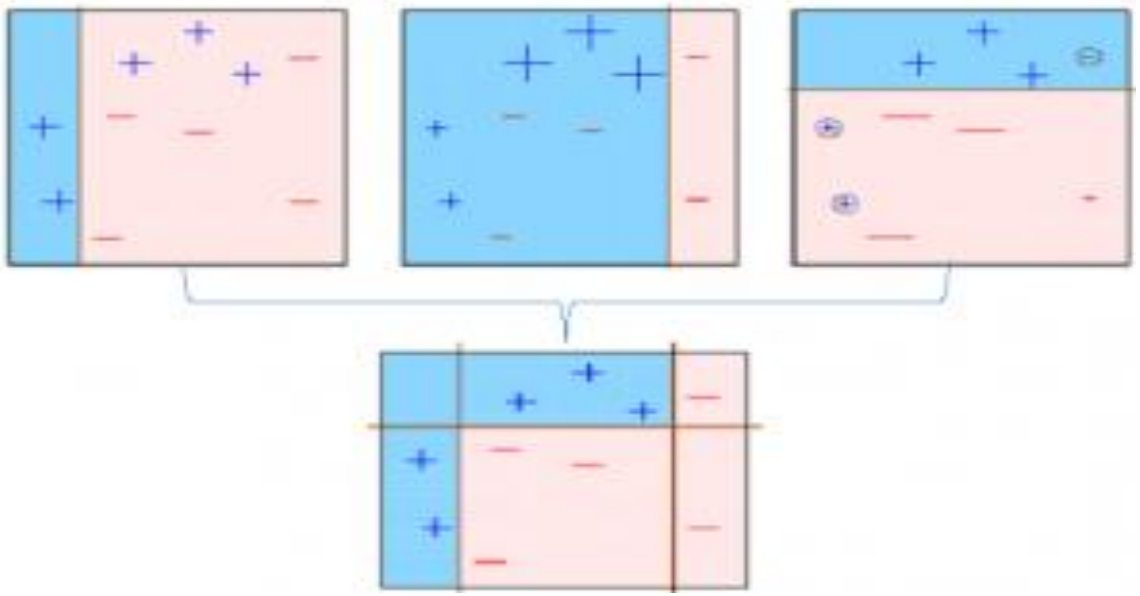
(This model tries to correct the errors from the previous model)



8.Similarly, multiple models are created, each correcting the errors of the previous model.

9.The final model (strong learner) is the weighted mean of all the models (weak learners)

# Boosting



Thus, the boosting algorithm combines a number of weak learners to form a strong learner.

The individual models would not perform well on the entire dataset, but they work well for some part of the dataset.

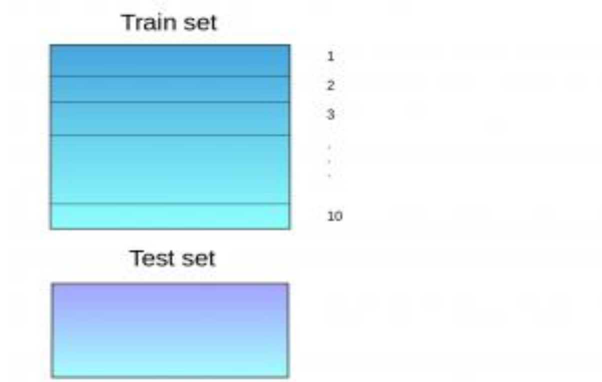
Thus, each model actually boosts the performance of the ensemble

# Stacking

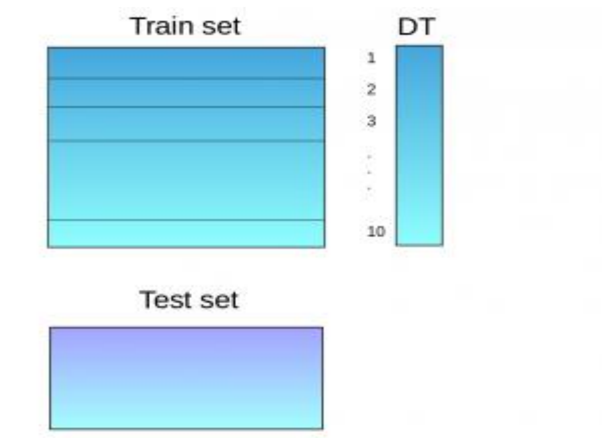
**Stacking is an ensemble learning technique that uses predictions from multiple models (for example decision tree, knn or svm) to build a new model.**

This model is used for making predictions on the test set.

1. Below is a step-wise explanation for a simple stacked ensemble: The train set is split into 10 parts-



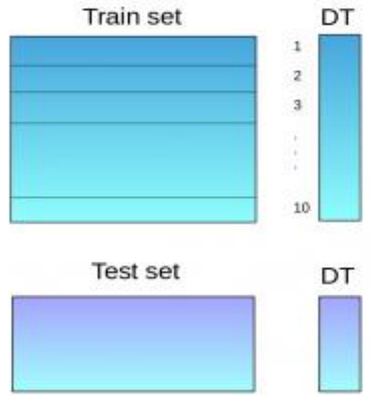

2. A base model (suppose a decision tree) is fitted on 9 parts and predictions are made for the 10th part. This is done for each part of the train set.



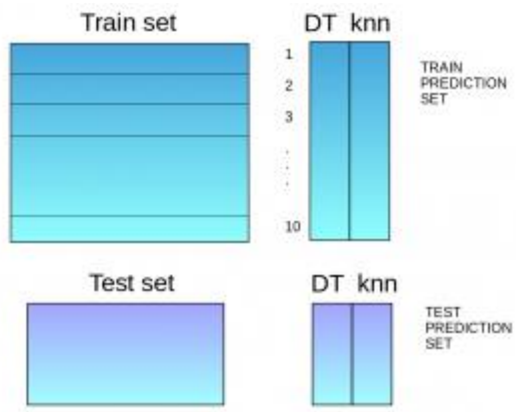
# Stacking

3. The base model (in this case, decision tree) is then fitted on the whole train dataset.

4.Using this model, predictions are made on the test set.

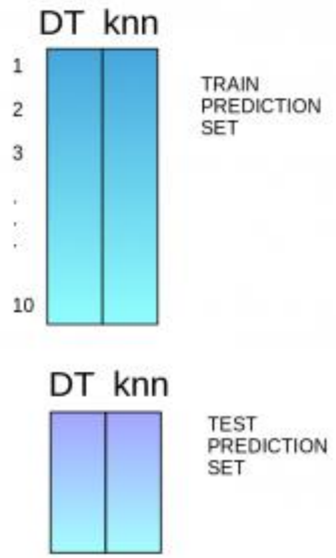



5. Steps 2 to 4 are repeated for another base model (say knn) resulting in another set of predictions for the train set and test set.



# Stacking

6. The predictions from the train set are used as features to build a new model.




7. This model is used to make final predictions on the test prediction set.

# Algorithms based on Bagging and Boosting

**Bagging algorithms:**

1. Bagging meta-estimator
2. Random forest

**Boosting algorithms:**

1. AdaBoost
2. GBM
3. XGBM
4. Light GBM
5. CatBoost

# Summary

- Boosting combines many weak classifiers to produce a powerful committee.
- It comes with a set of theoretical guarantee (e.g., training error, test error)
- It performs well on many tasks.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